Ethanol Blending in Gasoline - Bolivia

August 2025

U.S. GRAINS & BIOPRODUCTS COUNCIL

20 F St. NW, Suite 900 \ Washington, DC 20001
202-789-0789 \ www.grains.org

FARO90

www.faro90.com
+52 55 6122 2255

Ethanol Blending in Latin America

There are important fuel quality and environmental impact of vehicle emission challenges in the Region.

- The use of ethanol improves gasoline quality and creates flexibility in gasoline production.
- Ethanol use is a cost-effective way to increase gasoline octane and to replace more expensive gasoline components.
- Ethanol contributes to transport decarbonization and air quality improvement.
- There are opportunities across Latin America to increase the ethanol blend level and implement new policies on the use of gasoline-ethanol blends.

Sixteen countries with potential and additional use of ethanol were studied: 1) gasoline market profiles; 2) Optimization of gasoline blends with ethanol and 3) Environmental impact of gasolines blended with ethanol.

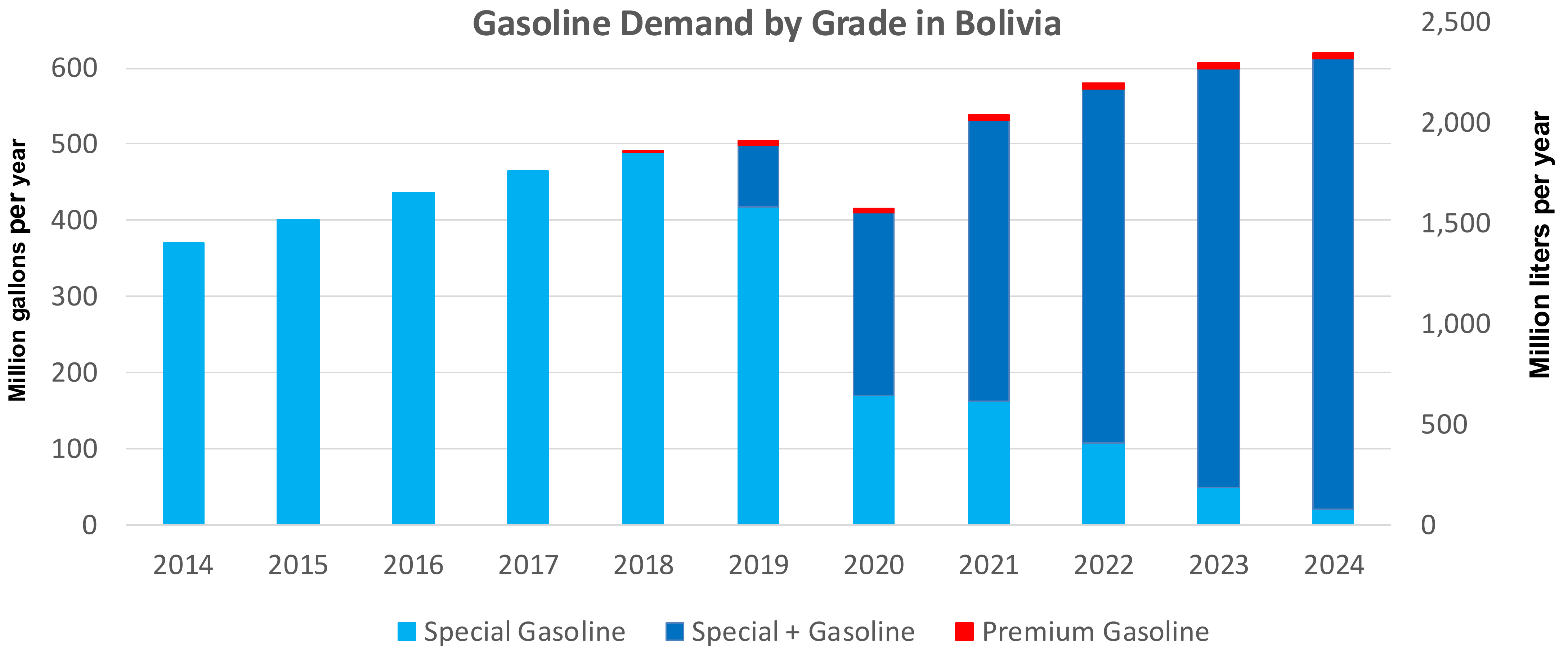
Ethanol Blending in Gasoline - Bolivia

In 2024, gasoline demand was 620 million gallons (2,348 million liters). Gasoline Imports represented 51.4% of demand, and are supplied from Brazil, Chile, Argentina, and the United States. Regular Gasoline (Especial) with 85 RON Octane represented 96% of demand and Premium Gasoline with a 95 RON octane the rest. Gasoline production has decreased in the last years.

Gasoline E8 and E12 are currently marketed (Decreto Supremo No. 4718:ANH). About 80% of gasoline demand contains ethanol. All ethanol consumed in Bolivia is produced by five local companies: Azúcar Aguaí, Unagro, Poplar Capital, Granosol, and Guabirá.

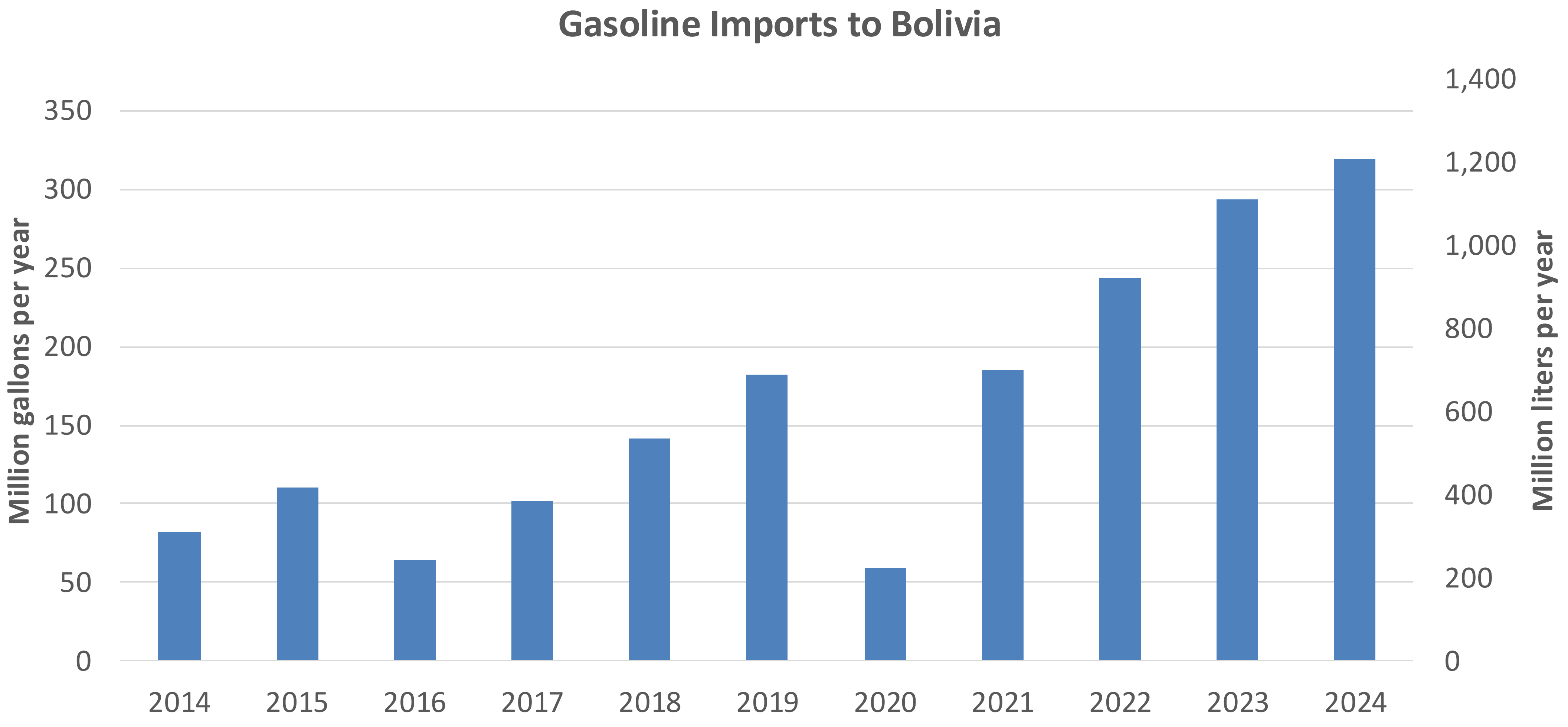
Source: ANH- Bolivia

Gasoline Demand in Bolivia



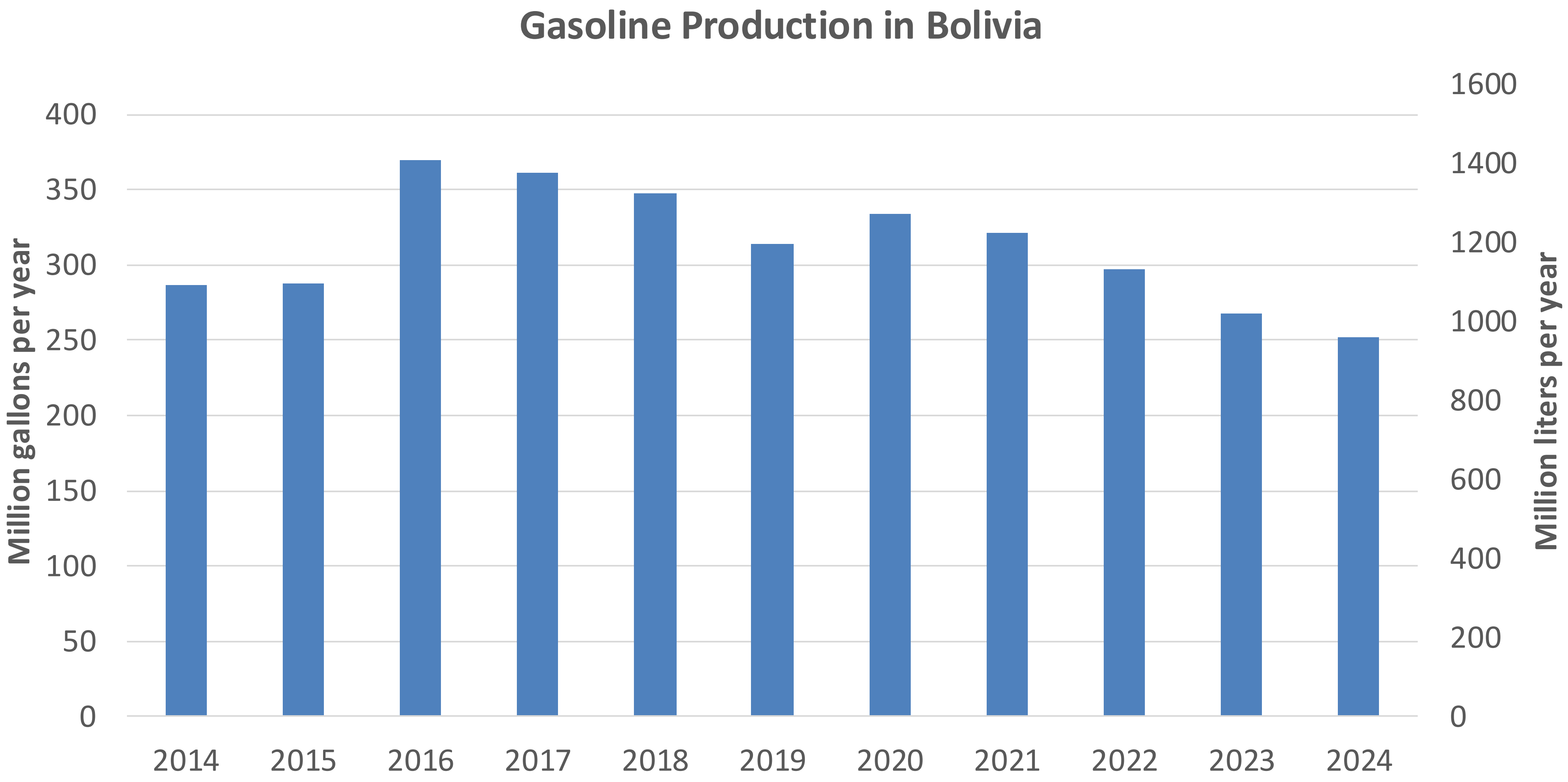
Source: ANH,INE - Bolivia

Gasoline Imports in Bolivia



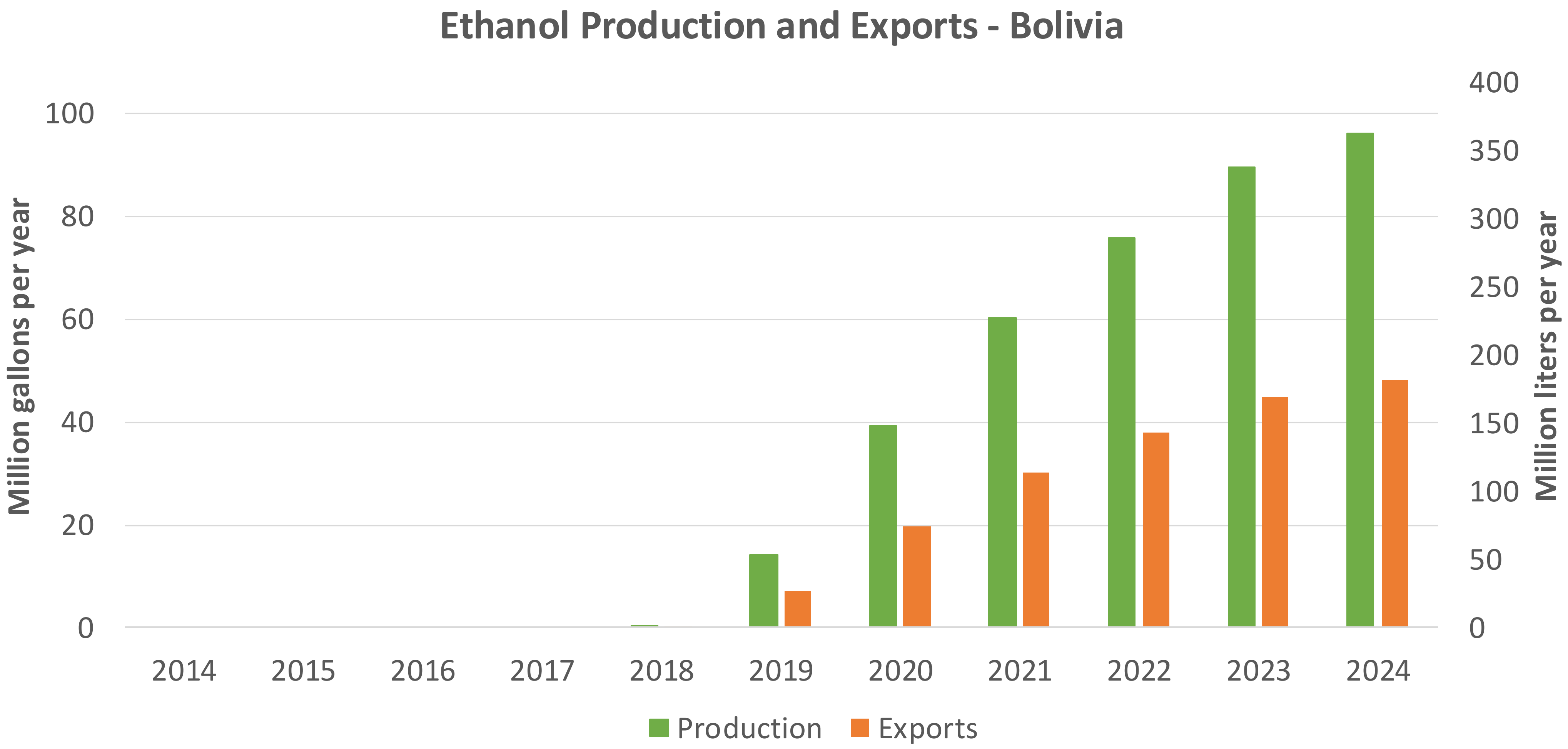
Source: ANH,INE - Bolivia

Gasoline Production in Bolivia



Source: ANH,INE - Bolivia

Ethanol Production in Bolivia



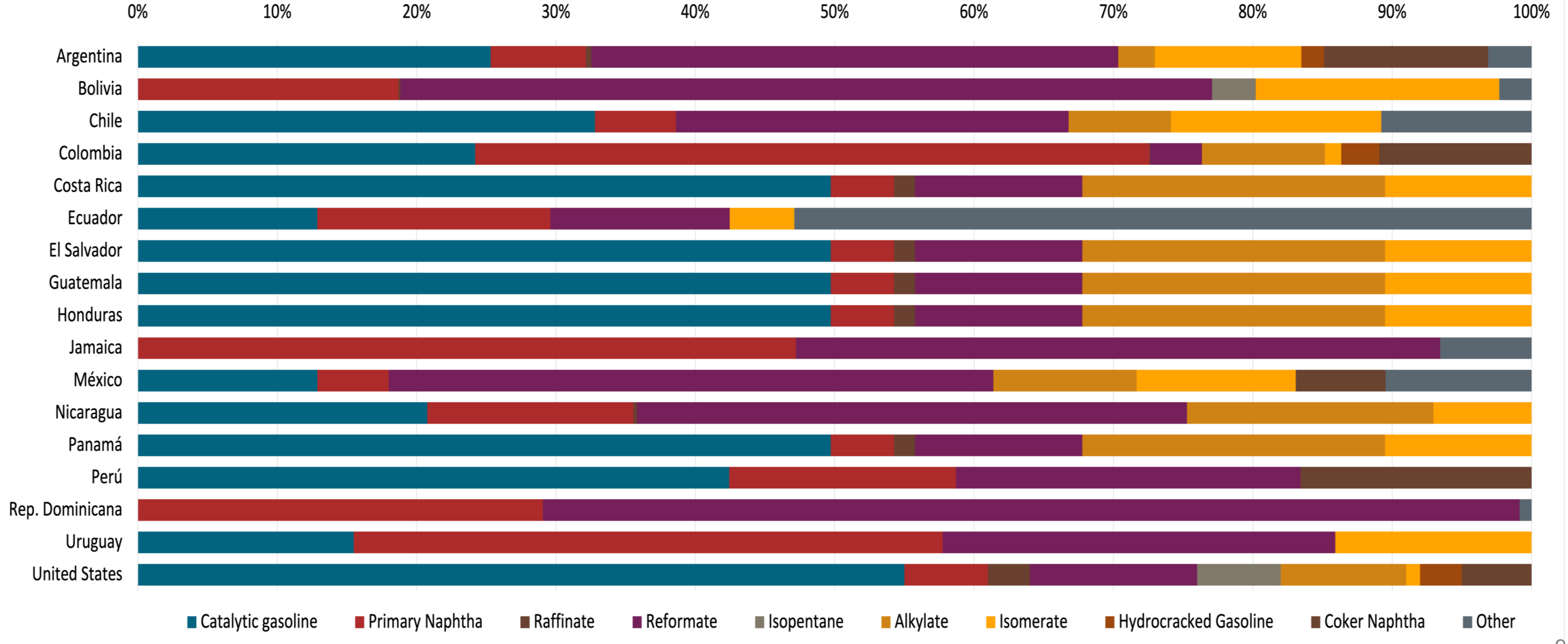
Source: ANH,INE - Bolivia

Gasoline Quality in Bolivia

Name	Supreme Decree 4718		EN 228:2012 + A1:2017 (Euro 6 enabling)			
Implementation Date	2022		2017			
Applicability	Whole country	Whole country	All countries			
Selected Grade	Gasoline Regular	Gasoline Premium	RON 95 E5	RON 95 E10	RON 98 E5	RON 98 E10
Benzene Content	< 3%v/v (2,7%v/v for E12 in summer)	< 3%v/v (2,7%v/v for E12 in summer)	< 1 %v/v	< 1 %v/v	< 1 %v/v	< 1 %v/v
Aromatics	< 42 % v/v	< 48 %v/v	< 35 %v/v	< 35 %v/v	< 35 %v/v	< 35 %v/v
Olefins	< 18 %v/v	< 18 %v/v	< 18 %v/v	< 18 %v/v	< 18 %v/v	< 18 %v/v
Lead Content	< 0,013 g/l	< 0,013 g/l	< 5 mg/l	< 5 mg/l	< 5 mg/l	< 5 mg/l
Manganese	< 18 mg/l	< 18 mg/l	< 2,0 mg/l	< 2,0 mg/l	< 2,0 mg/l	< 2,0 mg/l
RON	> 85	> 95	> 95	> 95	> 98	> 98
MON	-	-	> 85	> 88	> 85	> 88
AKI						
Sulfur Content	< 500 mg/kg	< 500 mg/kg	< 10 mg/kg	< 10 mg/kg	< 10 mg/kg	< 10 mg/kg
Oxygen Content	< 2,7 %m/m	< 2,7 %m/m	<2,7 % m/m	<3,7 % m/m	<2,7 % m/m	<3,7 % m/m
Ethanol (EtOH)	< 12 %v/v	< 12 %v/v	<5 %v/v	<10 %v/v	<5 %v/v	<10 %v/v
RVP 37.8°C (Summer)	<>7-9 psi	<> 7-9 psi	<> 60 - 70 kPa *Depends on the country, RVP is regulated in the EU Fuel Quality Directive			
RVP 37.8 °C(Winter)	<>7-9.5 psi	<> 7-9.5 psi				
RVP 37.8°C (Transition)	<>7-9 psi	<> 7-9 psi				
MTBE			-	-	-	-
Ethers 5 or more C Atoms	-	-	Based on oxygen content	<22 %v/v	Based on oxygen content	<22 %v/v

Gasoline Component Blending in Latin America

Gasoline is a blend of a base gasoline and other components. This blending is usually done at blending terminals as only 30% of the world's finished gasoline is distributed directly from refineries. Each component provides different properties to the final blend, for example, isomerates, alkylates and butanes increase the octane. The components commonly used in Latin America are:



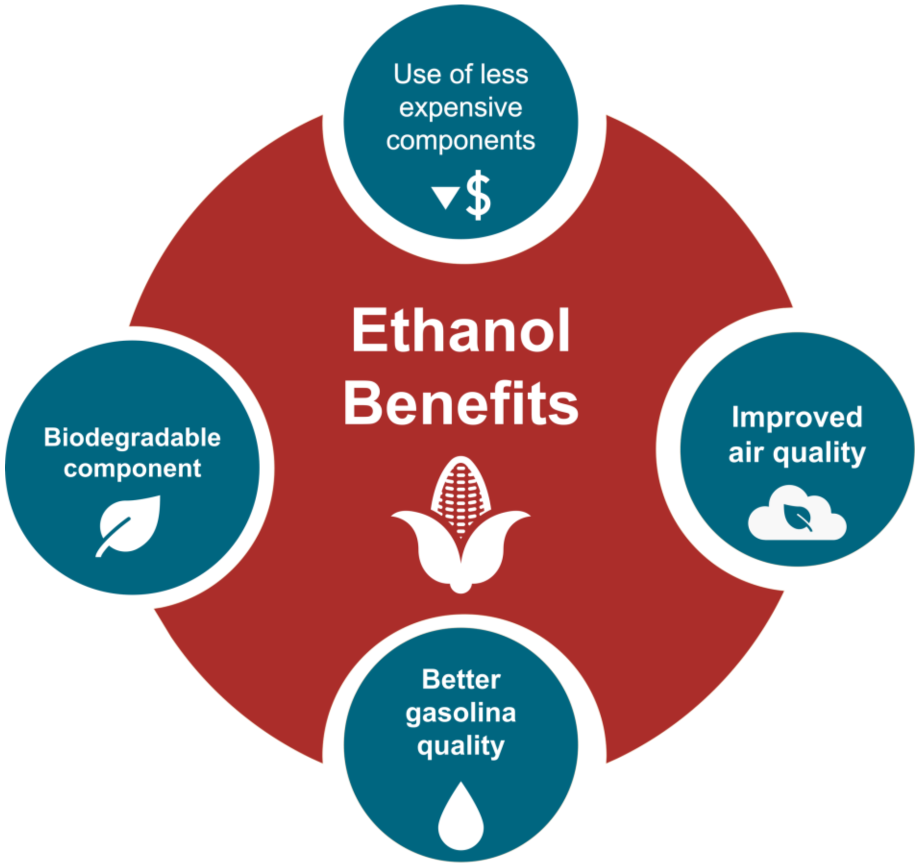
Gasoline Blending Optimization

In some parts of the world, ethanol is added to gasoline as a blending component. The advantages of ethanol include that it is a renewable fuel made of biomass; that it is an octane booster that helps to dilute sulfur; and that it allows the fulfillment of environmental objectives. To determine the optimal components to be blended with ethanol, a **blending model** was used. This model selects the components to add in the gasoline/ethanol blend based on:

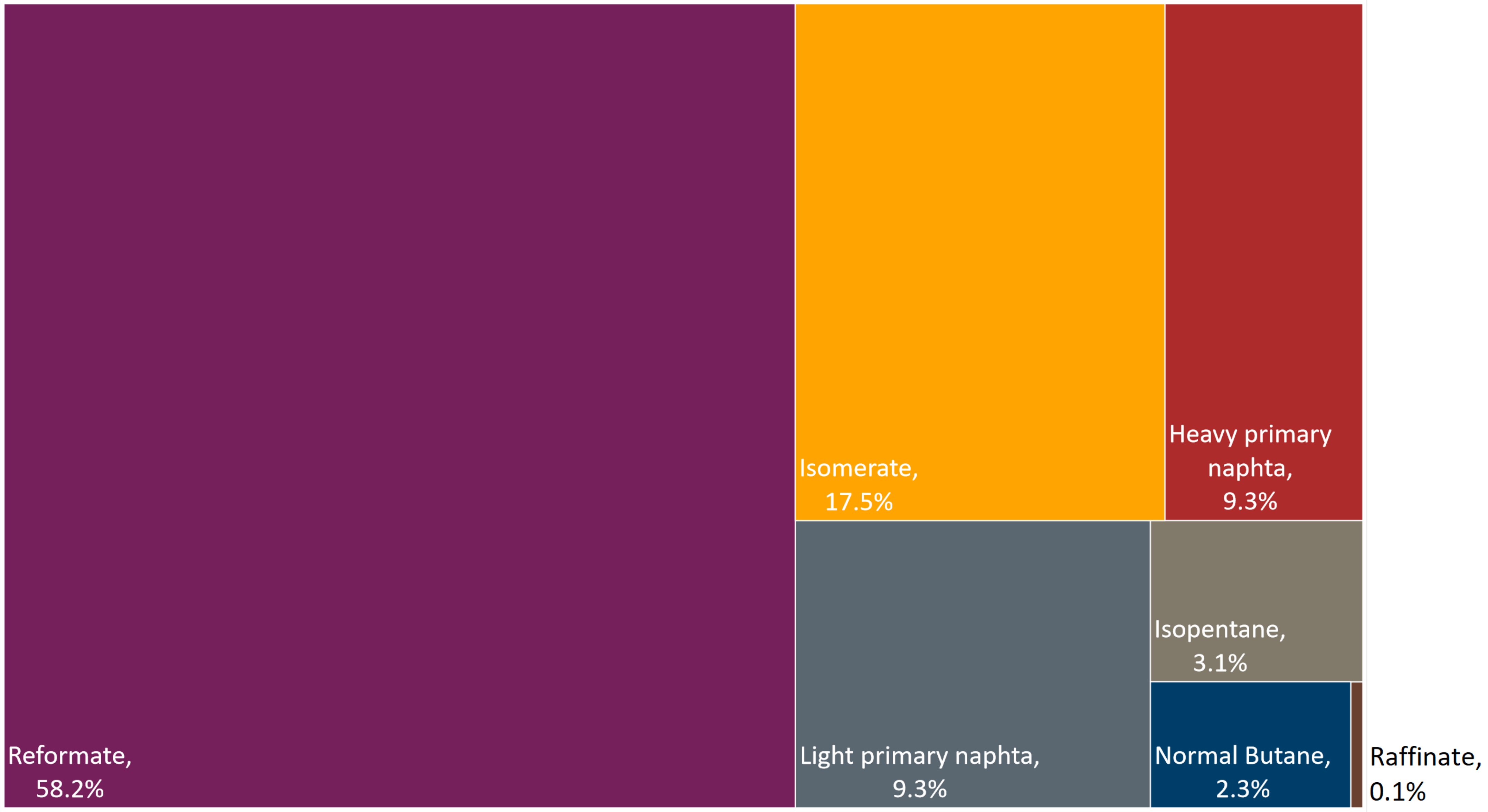
- Components prices,
- Properties each component affects,
- Quality parameters by country, and
- Component availability by country.

Through iterations, the model obtains the %v/v of the components to be blended with 10%, 15%, 20%, 25% and 30% of ethanol, in such a way that the final blend complies with the required properties of a finished gasoline by country.

The blending model uses gasoline component spot average 2024 prices and provides fuel prices that do not include country distribution costs, local taxes and subsidies and import or gas station margins.

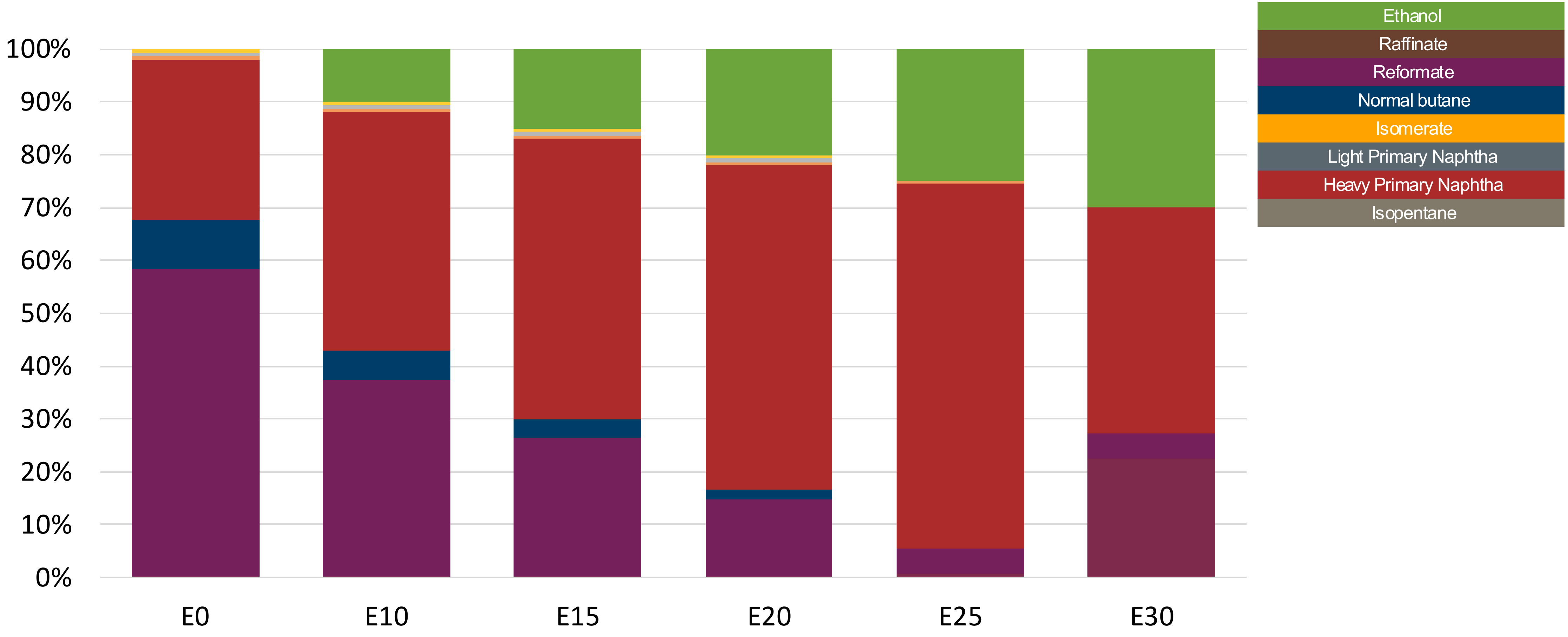


Available Blending Components



Source: Faro90

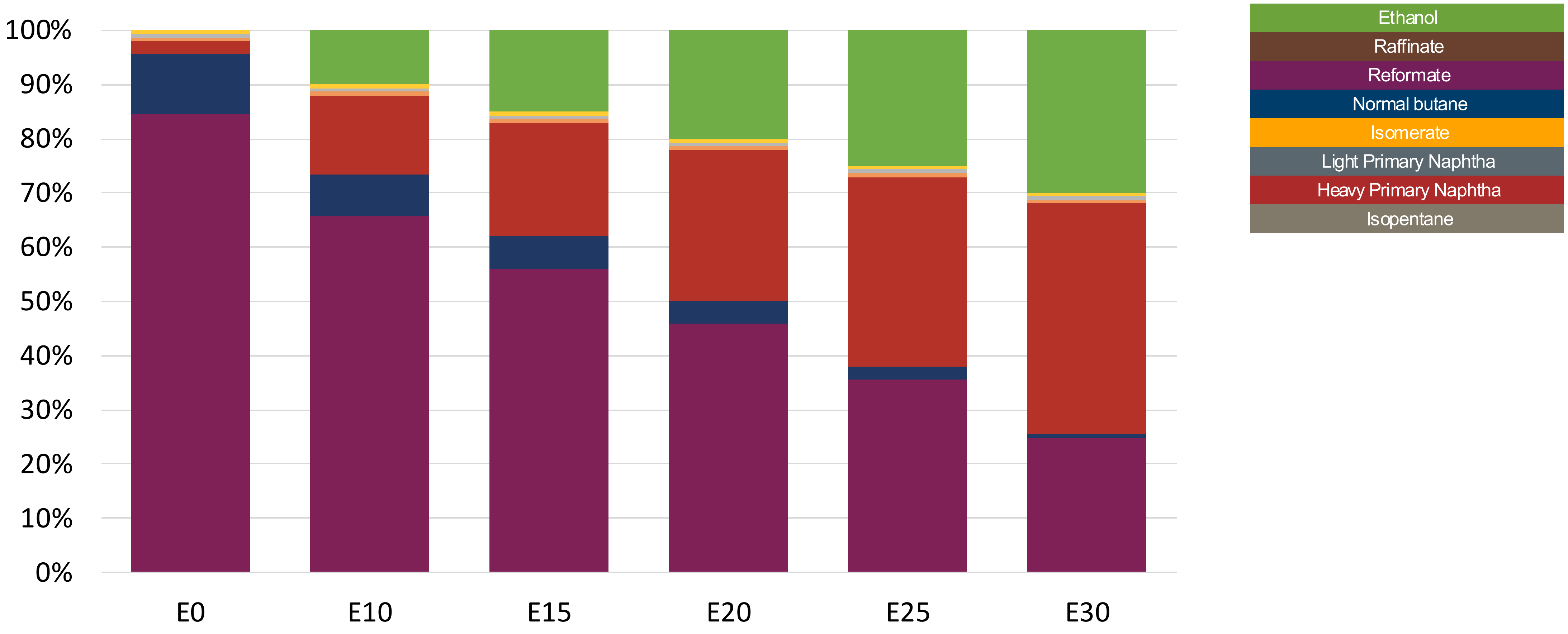
Ethanol Blending - Gasoline Especial – Constant Octane



Octane (RON)	85.0	85.0	85.0	85.0	85.0	89.2
Price (US$/gal)	$2.26	$1.96	$1.79	$1.62	$1.44	$1.54

Source: Faro90

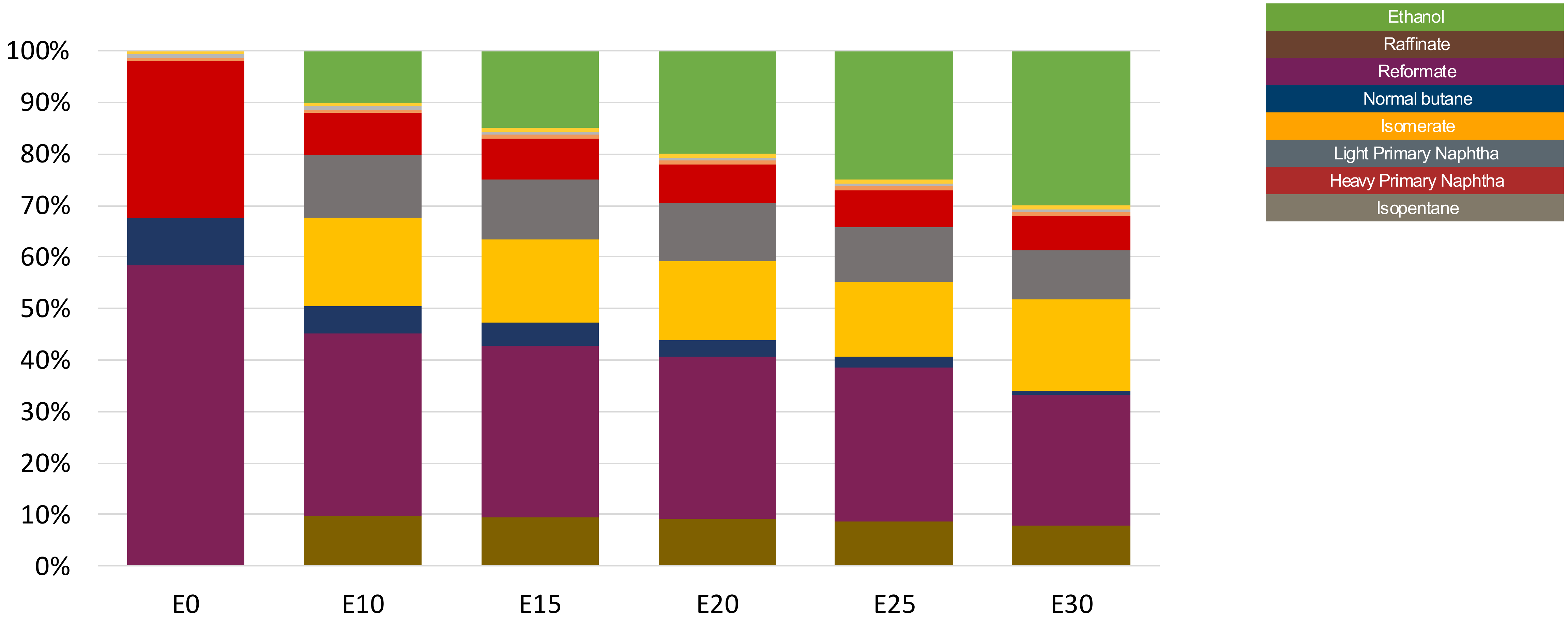
Ethanol Blending – Gasoline Premium – Constant Octane



Octane (RON)	95.0	95.0	95.0	95.0	95.0	95.0
Price (US$/gal)	$2.76	$2.50	$2.37	$2.22	$2.08	$1.91

Source: Faro90

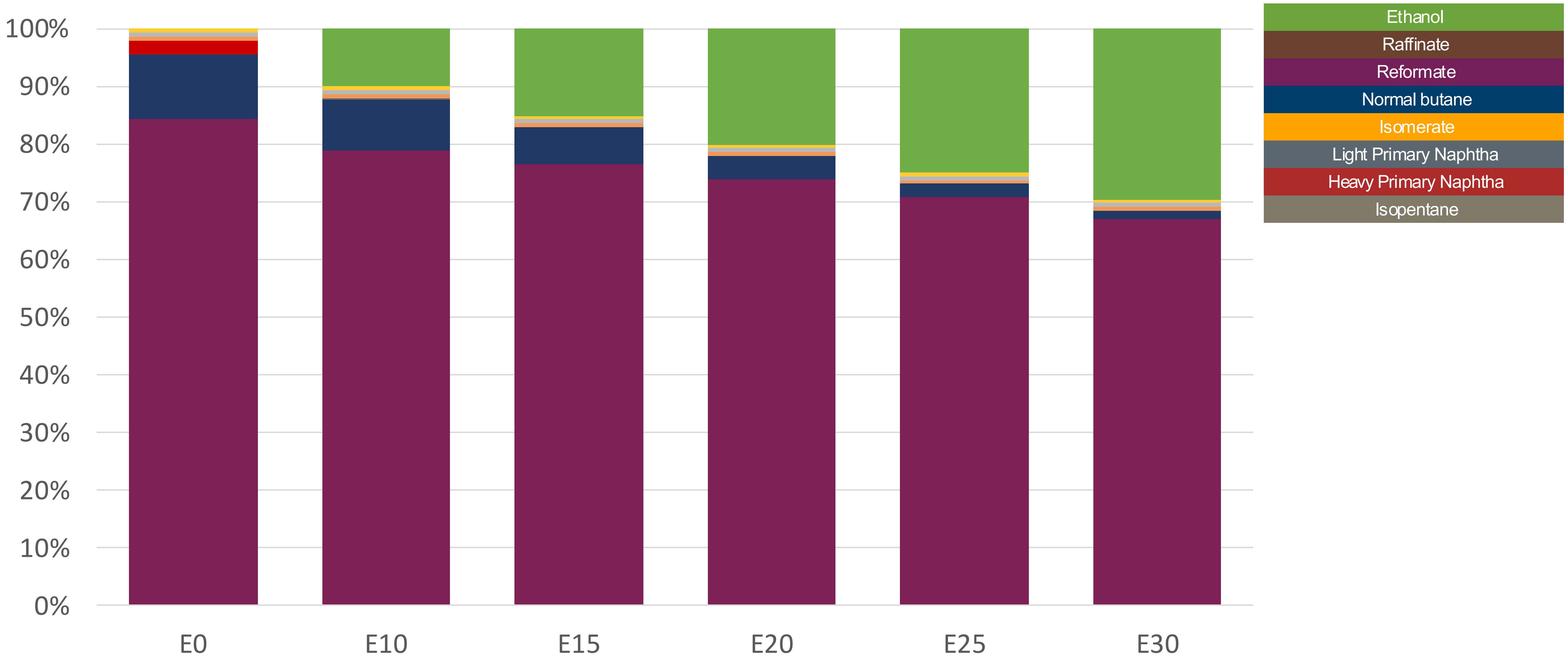
Ethanol Blending - Gasoline Especial – Octane Increment



Octane (RON)	85.0	90.2	92.8	95.3	97.7	99.9
Price (US$/gal)	$2.26	$2.26	$2.26	$2.26	$2.26	$2.26

Source: Faro90

Ethanol Blending - Gasoline Premium – Octane Increment



	E0	E10	E15	E20	E25	E30
Octane (RON)	95.0	99.7	101.6	103.4	105.4	107.7
Price (US$/gal)	$2.76	$2.76	$2.76	$2.76	$2.76	$2.76

Source: Faro90

Vehicle Emission Impact for Ethanol Gasoline Blending

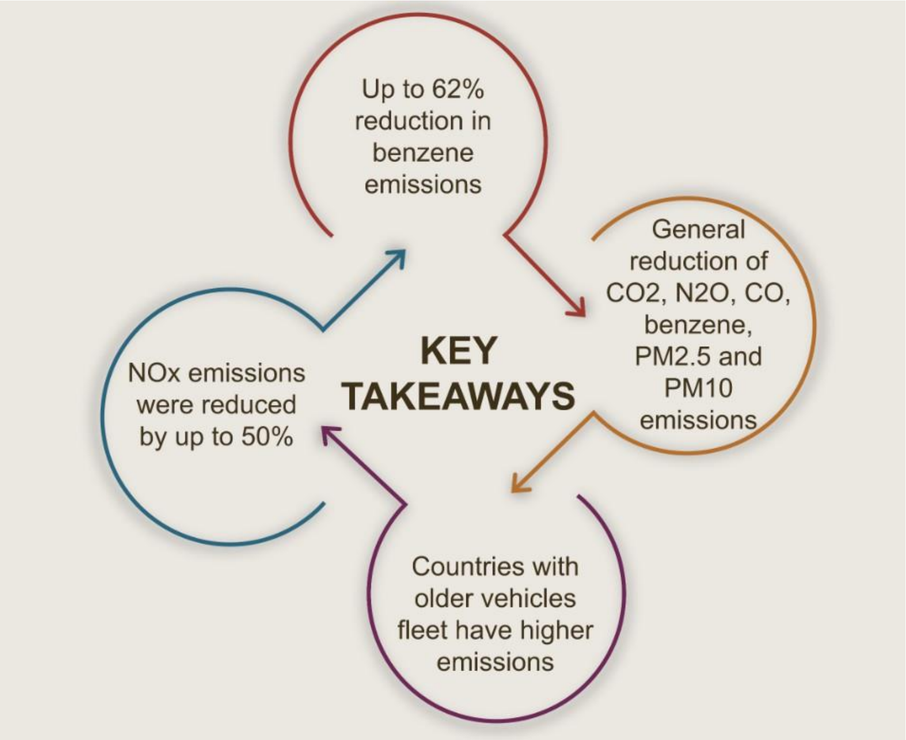
The model used in this analysis takes as a reference the **International Vehicle Emissions Model (IVE).**
The model uses the Base Emission Rates from IVE model, as well as its Adjustment Factors based on:

- Vehicle technology (cars, trucks, buses, motorcycles),
- Vehicle fleet average age,
- Average traveled distance per vehicle by country, as well as
- Geographical and climatic conditions (altitude, humidity, temperature).

Emissions of criteria pollutants, toxic pollutants, and greenhouse gases (GHG) were calculated and calibrated with emission inventories, using real gasoline quality data. The reduction rates for gasoline/ethanol blends were obtained from various sources (IPCC, US Grains, among others).

Emission estimations for different pollutants for gasoline and gasoline/ethanol blends (10%, 15%, 20%, 25% and 30% ethanol) were determined using the IVE Model. A comparison between the results and the European (Euro 6) requirements is made. Results are also compared with real emissions of the United States vehicle fleet*.

Main Results



**Source: Bureau of transportation statistics.*

Gasoline Vehicle Fleet Bolivia

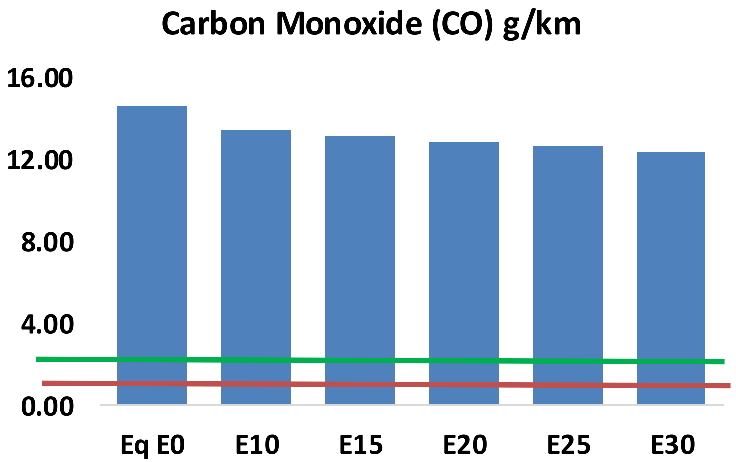
Vehicle Type	Motor	Tailpipe	Recuperator	Age	Number of vehicles
Auto/Sml Truck	Multi FI-Pt	Euro V	PCV Tank	>4years, >80 mkm	39,474
				4-8 years, 80-160 mkm	56,957
				<8 years, < 160 mkm	39,689
		Euro IV		<8 years, < 160 mkm	43,498
		Euro III		<8 years, < 160 mkm	24,308
		3-Ways		<8 years, < 160 mkm	249,368
Sml Engine	FI 4-cycle	Euro III	PCV	>4years, >80 mkm	258,749
				4-8 years, 80-160 mkm	89,752
		Euro II		4-8 years, 80-160 mkm	89,752
				<8 years, < 160 mkm	440,804
Truck	FI	Euro V	PCV	>4years, >80 mkm	178,544
				4-8 years, 80-160 mkm	203,281
				<8 years, < 160 mkm	135,905
		Euro IV		<8 years, < 160 mkm	132,960
		Euro III		<8 years, < 160 mkm	126,191
		3-Ways		<8 years, < 160 mkm	280,182

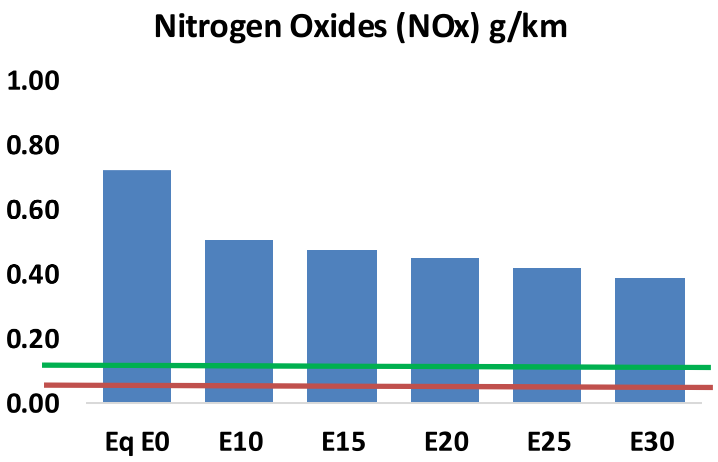
Vehicular Fleet: **2,389,413** Average Age:
12 years Motorcycles**: 36.8%**

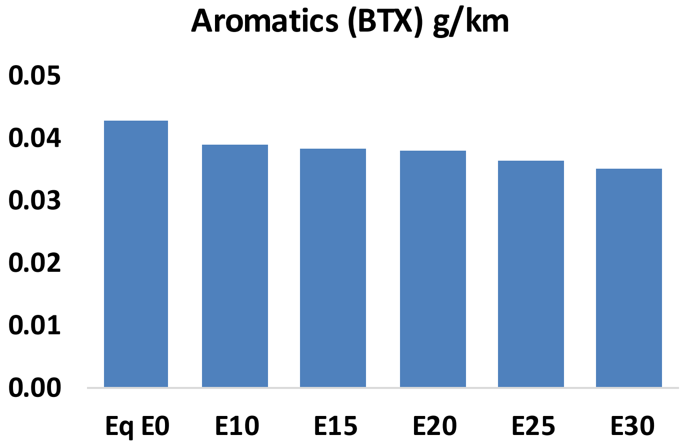
Source: Instituto Nacional de Estadística – Bolivia (INE), analysis Faro 90

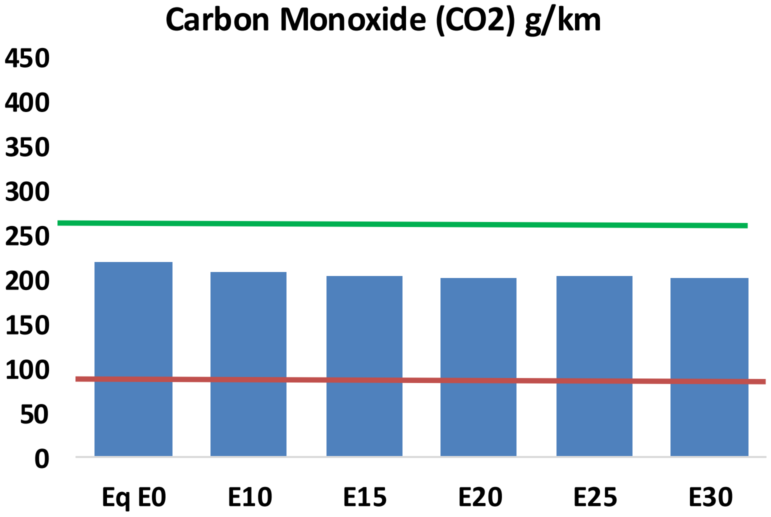
Bolivia – Vehicular Emissions

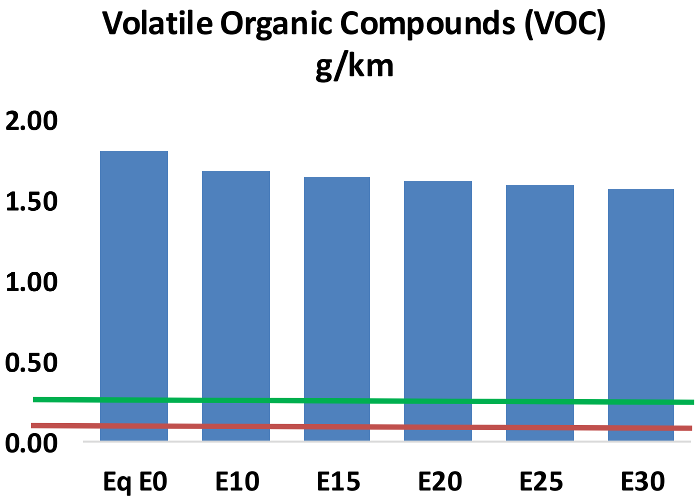
United States
Euro 6

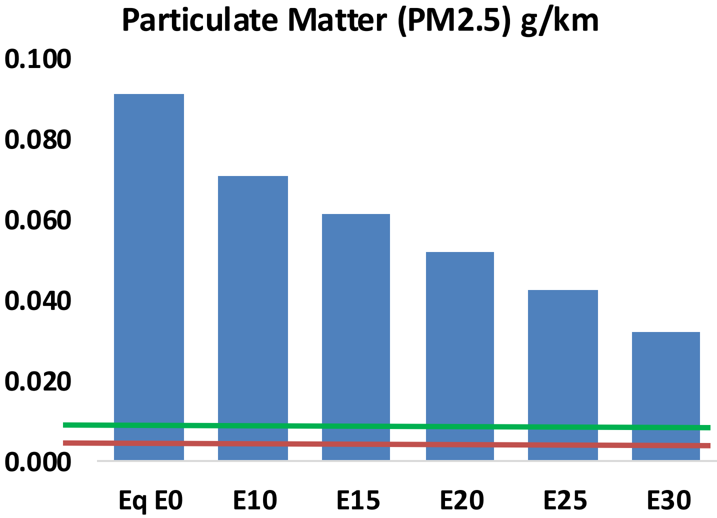












Source: Faro90

Bolivia – Vehicular Emissions

Vehicular Fleet Emissions (kg/day)	Eq E0	E5	E10	E15	E20	E25	E30	Reduction (%)	
								E10	E20
CO	675,984	650,708	625,432	609,927	596,722	587,291	572,152	-7%	-12%
CO2	10,167,733	9,906,277	9,659,346	9,465,142	9,369,043	9,514,488	9,339,109	-5%	-8%
VOC	83,744	80,916	78,087	76,473	75,173	74,271	72,722	-7%	-10%
NOx	33,437	25,078	23,406	22,060	20,805	19,420	17,956	-30%	-38%
SOx	132	122	112	103	95	86	77	-15%	-28%
NH3	2,921	2,892	2,863	2,877	2,909	2,932	2,951	-2%	0%
N2O	996	991	987	992	1,012	1,024	1,035	-1%	2%
CH4	18,986	18,986	18,986	19,271	19,689	20,049	20,391	0%	4%
Lead	-	-	-	-	-	-	-	0%	0%
Butadiene	353	337	322	312	302	296	285	-9%	-14%
Acetaldehyde	1,118	1,284	1,884	2,868	3,908	4,465	5,276	68%	249%
Formaldehyde	4,768	4,636	5,404	6,345	6,648	7,354	8,006	13%	39%
Benzene	1,981	1,903	1,804	1,775	1,760	1,683	1,629	-9%	-11%
PM2.5	862	765	669	581	493	400	303	-22%	-43%
PM10	3,354	2,978	2,602	2,260	1,919	1,557	1,180	-22%	-43%

Source: Faro90

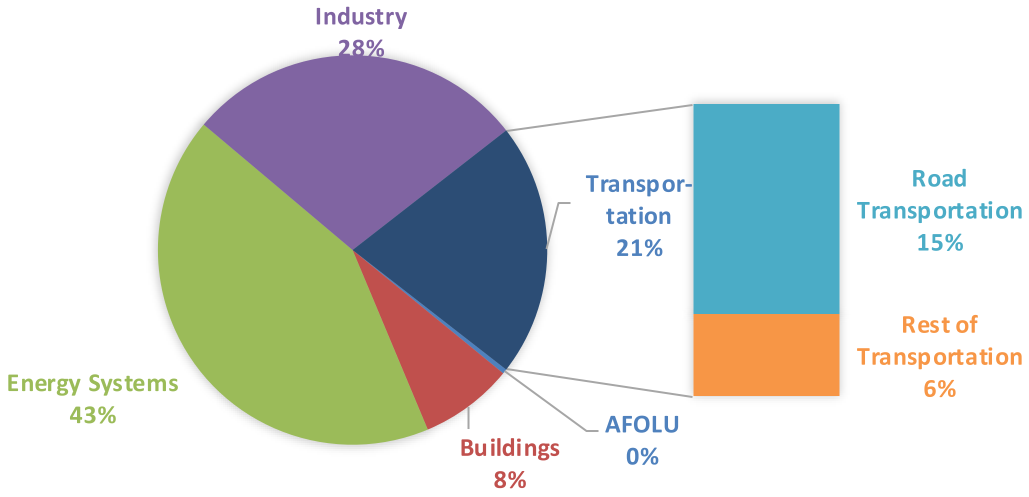
Life Cycle GHG Emissions

The Global Agenda for Climate Action calls for countries, cities and regions, businesses, and members of civil society around the world to to achieve a low-carbon and climate-resilient economies in support of the Paris Agreement. Road Transportation accounts for 15% of total GHG emissions (IPCC,2023).

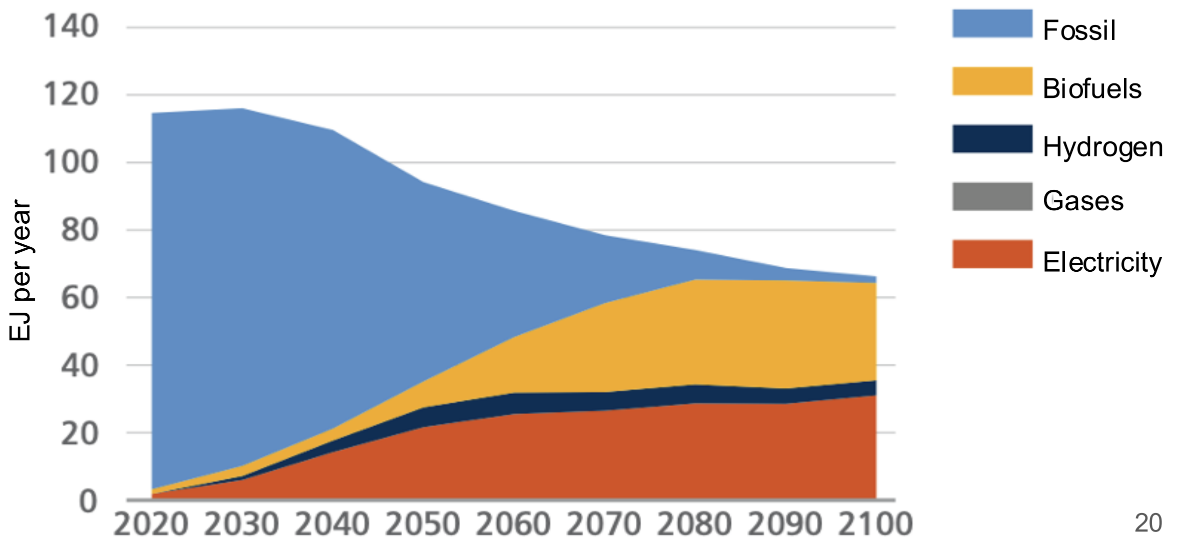
Ethanol gasoline blends are an internationally adopted practice that offer significant short-term benefits. Biofuels are an important solution to achieve reduction targets in the transportation sector.

Country GHG emission reductions are calculated using **RED II/III** and **GREET** life cycle emission methodologies and the **International Vehicular Emissions Model (IVE).** Results are compared to the current country targets for 2035. Current 2024 country emission are those published by **IPCC**.

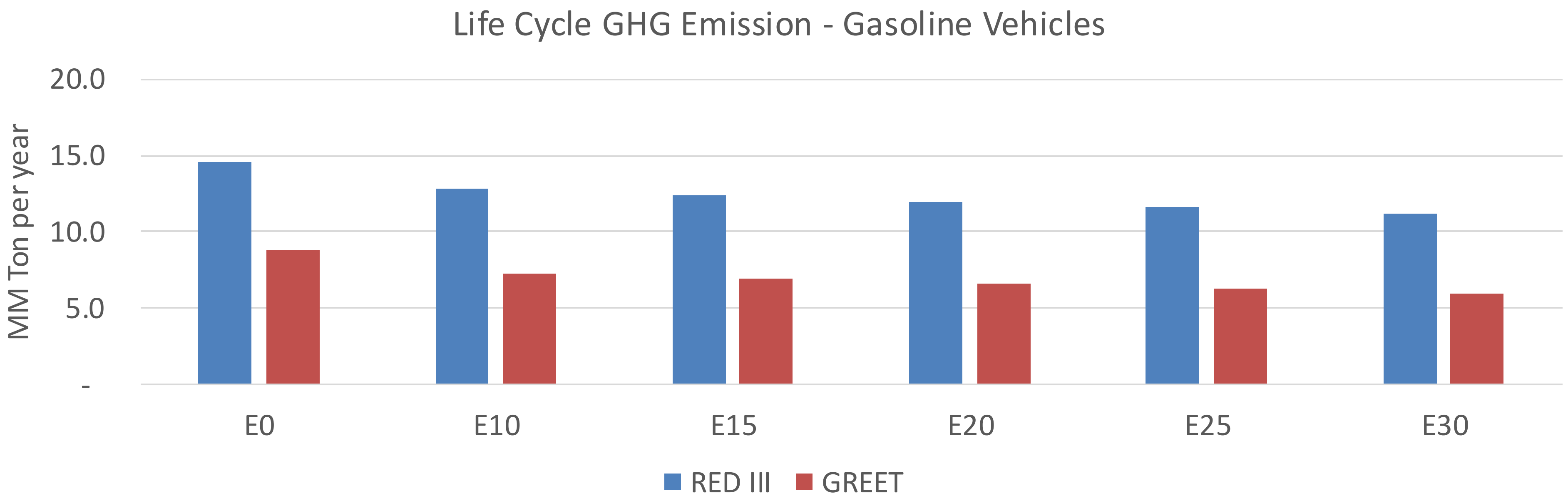
Current GHG Global Emissions Distribution



IPCC Transport Sector Sustainable Development Goals and climate policies Scenario, 2023



Bolivia – Gasoline Vehicle Fleet GEI Emissions



Country Emissions 2024 (MMT/year)	63.8
2035 Target (MMT/year)	44.6
Delta	19.1

%Reduction vs E0	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	17.7%	21.8%	25.4%	28.2%	32.5%
RED II/III	0.0%	12.1%	15.3%	18.1%	20.4%	23.6%

% 2035 Target	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	8.2%	10.0%	11.7%	13.0%	15.0%
RED II/III	0.0%	9.2%	11.7%	13.8%	15.6%	18.0%

Source: Greet, RED II/III, climateactiontracker.org, F90